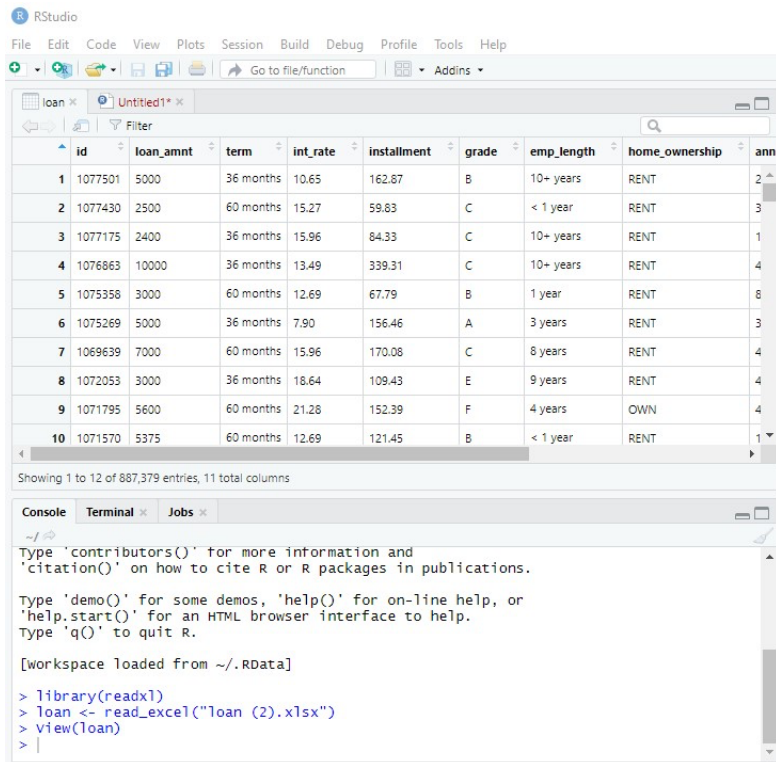


Week 5 Sean Graham

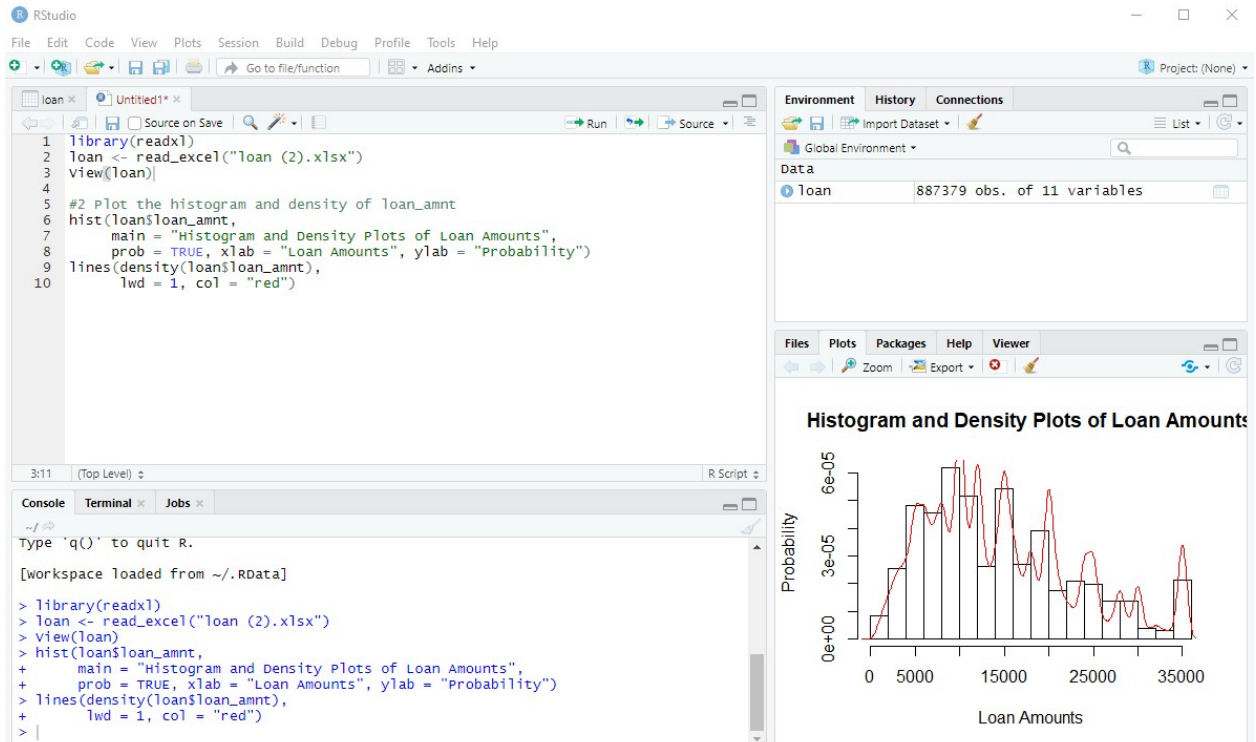
1.



RStudio interface showing a data frame 'loan' with 11 columns and 887,379 rows. The columns are: id, loan_amnt, term, int_rate, installment, grade, emp_length, home_ownership, and annuity. The console shows the R code used to load the data:

```
[workspace loaded from ~/.RData]
> library(readxl)
> loan <- read_excel("loan (2).xlsx")
> view(loan)
```

2.

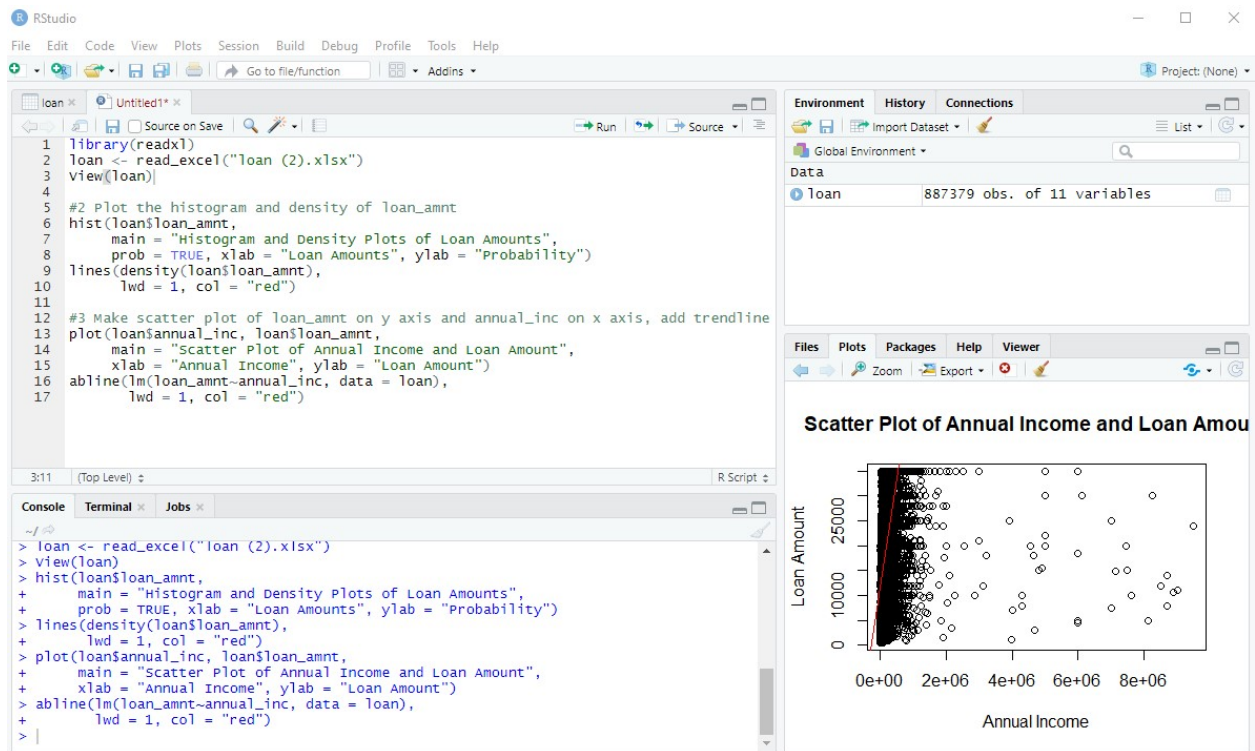


RStudio interface showing the R code used to create a histogram and density plot of loan amounts. The Environment pane shows the 'loan' data frame with 887,379 observations. The Plots pane shows the resulting histogram and density plot.

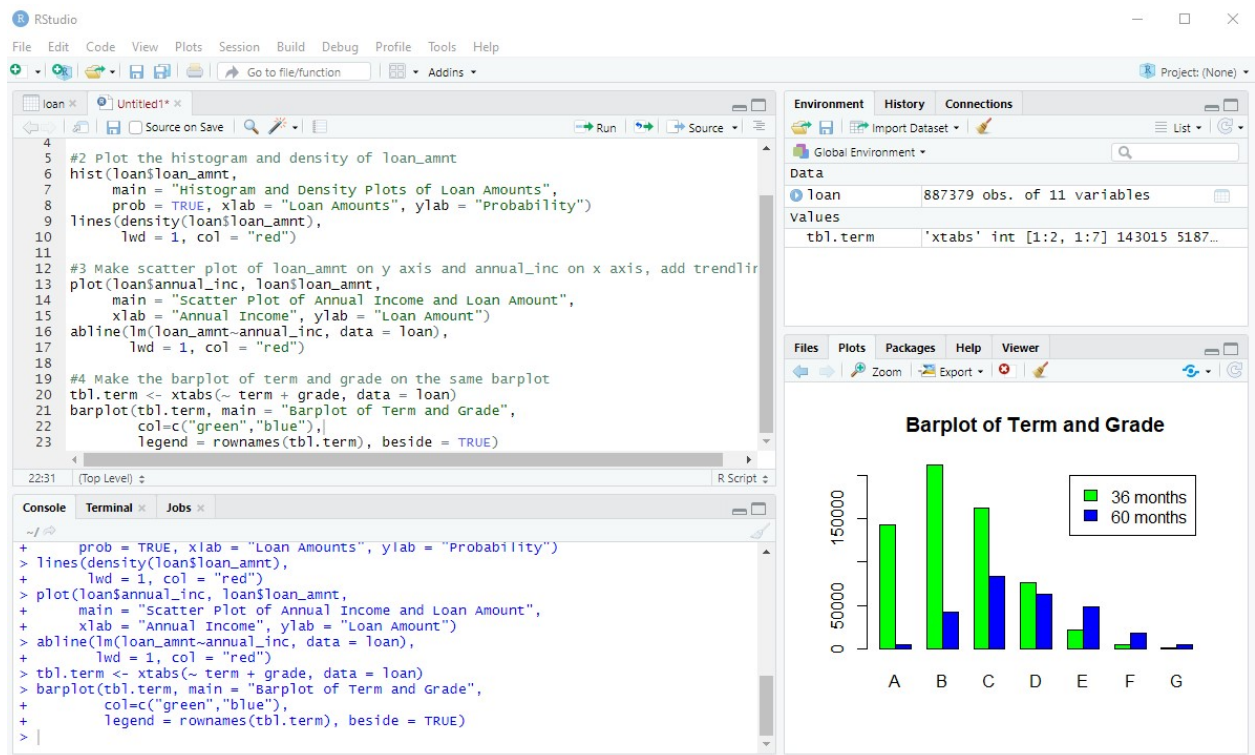
```
1 library(readxl)
2 loan <- read_excel("loan (2).xlsx")
3 view(loan)
4
5 #2 Plot the histogram and density of loan_amnt
6 hist(loan$loan_amnt,
7     main = "Histogram and Density Plots of Loan Amounts",
8     prob = TRUE, xlab = "Loan Amounts", ylab = "Probability")
9 lines(density(loan$loan_amnt),
10     lwd = 1, col = "red")
```

The histogram and density plot of loan amounts is displayed. The x-axis is labeled 'Loan Amounts' and ranges from 0 to 35,000. The y-axis is labeled 'Probability' and ranges from 0e+00 to 6e-05. The plot shows a distribution of loan amounts with a peak around 10,000 and a long tail extending to the right.

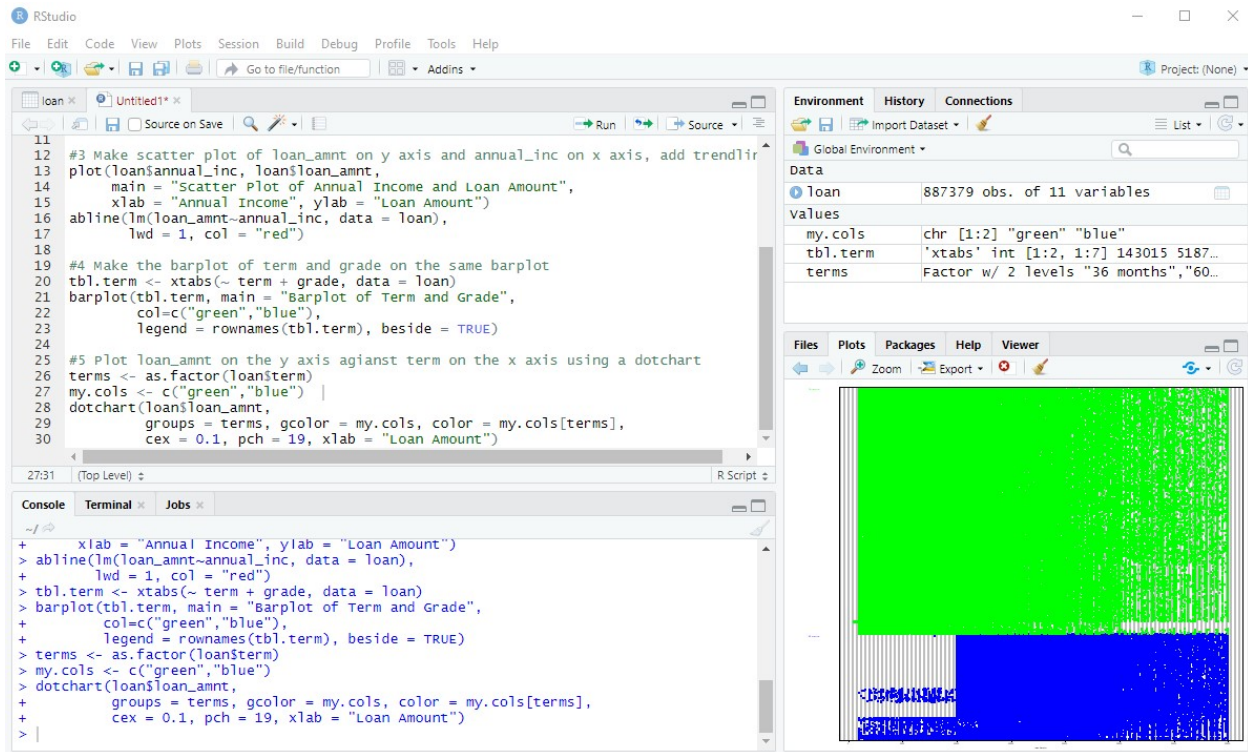
3.



4.



5. The axes and labels are there, they're just very small due to the large size of the chart.



6.

